

Thus any U_n can be obtained by finitely many applications of T_2 , S , or their inverses. Since f is invariant under T_2 and S , it is also invariant under U_m . So we find that

$$f(\mu^{-1}T_n\mu(\tau)) = f(\tau).$$

Therefore, if we let $F(\tau) = f(\mu^{-1}(\tau)) = f(1 - 1/\tau)$, we find that F is periodic of period 1, that is,

$$F(T_n\tau) = F(\tau) \quad \text{for every integer } n.$$

Now, by the previous argument, if we set $h(z) = F(\tau)$ with $z = e^{2\pi i\tau}$, we see that h has a removable singularity at $z = 0$, and the desired inequality follows by the maximum principle.

We conclude from this analysis that f attains its maximum in the interior of the upper half-plane, and this contradicts the maximum principle.

The proof of the two-squares theorem is now only one step away.

We consider the function $f(\tau) = \mathcal{C}(\tau)/\theta(\tau)^2$. Since we know by the product formula that $\theta(\tau)$ does not vanish in the upper half-plane (Corollary 1.4), we find that f is holomorphic in $\mathbb{H}$. Moreover, by Proposition 3.3, f is invariant under the transformations T_2 and S , that is, $f(\tau + 2) = f(\tau)$ and $f(-1/\tau) = f(\tau)$. Finally, in the fundamental domain $\mathcal{F}$, the function $f(\tau)$ is bounded, and in fact tends to 1 as $\text{Im}(\tau)$ tends to infinity, or as τ tends to the cusps ± 1 . This is because of properties (iii) and (iv) in Proposition 3.3, which are verified by both $\mathcal{C}$ and θ^2 . Thus f is bounded in $\mathbb{H}$. The result is that f is a constant, which must be 1, proving that $\theta(\tau)^2 = \mathcal{C}(\tau)$, and with it the two-squares theorem.

3.2 The four-squares theorem

Statement of the theorem

In the rest of this chapter, we shall consider the case of four squares. More precisely, we will prove that every positive integer is the sum of four squares, and moreover we will determine a formula for $r_4(n)$ that describes the number of ways this can be done.

We need to introduce another divisor function, which we denote by $\sigma_1^*(n)$, and which equals the sum of divisors of n that are not divisible by 4. The main theorem we shall prove is the following.

Theorem 3.6 *Every positive integer is the sum of four squares, and moreover*

$$r_4(n) = 8\sigma_1^*(n) \quad \text{for all } n \geq 1.$$

As before, we relate the sequence $\{r_4(n)\}$ via its generating function to an appropriate power of the function θ , which in this case is its fourth power. The result is that

$$\theta(\tau)^4 = \sum_{n=0}^{\infty} r_4(n)q^n$$

whenever $q = e^{\pi i \tau}$ with $\tau \in \mathbb{H}$.

The next step is to find the modular function whose equality with $\theta(\tau)^4$ expresses the identity $r_4(n) = 8\sigma_1^*(n)$. Unfortunately, here there is nothing as simple as the function $\mathcal{C}(\tau)$ that arose in the two-squares theorem; instead we shall need to construct a rather subtle variant of the Eisenstein series considered in the previous chapter. In fact, we define

$$E_2^*(\tau) = \sum_m \sum_n \frac{1}{\left(\frac{m\tau}{2} + n\right)^2} - \sum_m \sum_n \frac{1}{\left(m\tau + \frac{n}{2}\right)^2}$$

for $\tau \in \mathbb{H}$. The indicated order of summation is critical, since the above series do not converge absolutely. The following reduces the four-squares theorem to the modular properties of E_2^* .

Proposition 3.7 *The assertion $r_4(n) = 8\sigma_1^*(n)$ is equivalent to the identity*

$$\theta(\tau)^4 = \frac{-1}{\pi^2} E_2^*(\tau), \quad \text{where } \tau \in \mathbb{H}.$$

Proof. It suffices to prove that if $q = e^{\pi i \tau}$, then

$$\frac{-1}{\pi^2} E_2^*(\tau) = 1 + \sum_{k=1}^{\infty} 8\sigma_1^*(k)q^k.$$

First, recall the forbidden Eisenstein series that we considered in the last section of the previous chapter, and which is defined by

$$F(\tau) = \sum_m \left[\sum_n \frac{1}{(m\tau + n)^2} \right],$$

where the term $n = m = 0$ is omitted. Since the sum above is not absolutely convergent, the order of summation, first in n and then in m , is crucial. With this in mind, the definitions of E_2^* and F give immediately

$$(10) \quad E_2^*(\tau) = F\left(\frac{\tau}{2}\right) - 4F(2\tau).$$

In Corollary 2.6 (and Exercise 7) of the last chapter, we proved that

$$F(\tau) = \frac{\pi^2}{3} - 8\pi^2 \sum_{k=1}^{\infty} \sigma_1(k) e^{2\pi i k \tau},$$

where $\sigma_1(k)$ is the sum of the divisors of k .

Now observe that

$$\sigma_1^*(n) = \begin{cases} \sigma_1(n) & \text{if } n \text{ is not divisible by } 4, \\ \sigma_1(n) - 4\sigma_1(n/4) & \text{if } n \text{ is divisible by } 4. \end{cases}$$

Indeed, if n is not divisible by 4, then no divisors of n are divisible by 4. If $n = 4\tilde{n}$, and d is a divisor of n that is divisible by 4, say $d = 4\tilde{d}$, then $\tilde{d}$ divides $\tilde{n}$. This gives the second formula. Therefore, from this observation and (10) we find that

$$E_2^*(\tau) = -\pi^2 - 8\pi^2 \sum_{k=1}^{\infty} \sigma_1^*(k) e^{\pi i k \tau},$$

and the proof of the proposition is complete.

We have therefore reduced Theorem 3.6 to the identity $\theta^4 = -\pi^{-2}E_2^*$, and the key to establish this relation is that E_2^* satisfies the same modular properties as $\theta(\tau)^4$.

Proposition 3.8 *The function $E_2^*(\tau)$ defined in the upper half-plane has the following properties:*

- (i) $E_2^*(\tau + 2) = E_2^*(\tau)$.
- (ii) $E_2^*(\tau) = -\tau^{-2}E_2^*(-1/\tau)$.
- (iii) $E_2^*(\tau) \rightarrow -\pi^2$ as $\text{Im}(\tau) \rightarrow \infty$.
- (iv) $|E_2^*(1 - 1/\tau)| = O(|\tau^2 e^{\pi i \tau}|)$ as $\text{Im}(\tau) \rightarrow \infty$.

Moreover $-\pi^2\theta^4$ has the same properties.

The periodicity (i) of E_2^* is immediate from the definition. The proofs of the other properties of E_2^* are a little more involved.

Consider the forbidden Eisenstein series F and its reverse $\tilde{F}$, which is obtained from reversing the order of summation:

$$F(\tau) = \sum_m \sum_n \frac{1}{(m\tau + n)^2} \quad \text{and} \quad \tilde{F}(\tau) = \sum_n \sum_m \frac{1}{(m\tau + n)^2}.$$

In both cases, the term $n = m = 0$ is omitted.

Lemma 3.9 *The functions F and $\tilde{F}$ satisfy:*

- (a) $F(-1/\tau) = \tau^2 \tilde{F}(\tau)$,
- (b) $F(\tau) - \tilde{F}(\tau) = 2\pi i/\tau$,
- (c) $F(-1/\tau) = \tau^2 F(\tau) - 2\pi i\tau$.

Proof. Property (a) follows directly from the definitions of F and $\tilde{F}$, and the identity

$$(n + m(-1/\tau))^2 = \tau^{-2}(-m + n\tau)^2.$$

To prove property (b), we invoke the functional equation for the Dedekind eta function which was established earlier:

$$\eta(-1/\tau) = \sqrt{\tau/i} \eta(\tau),$$

where $\eta(\tau) = q^{1/12} \prod_{n=1}^{\infty} (1 - q^{2n})$, and $q = e^{\pi i \tau}$.

First, we take the logarithmic derivative of η with respect to the variable τ to find (by Proposition 3.2 in Chapter 5)

$$(\eta'/\eta)(\tau) = \frac{\pi i}{12} - 2\pi i \sum_{n=1}^{\infty} \frac{nq^{2n}}{1 - q^{2n}}.$$

However, if $\sigma_1(k)$ denotes the sum of the divisors of k , then one sees that

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{nq^{2n}}{1 - q^{2n}} &= \sum_{n=1}^{\infty} \sum_{\ell=0}^{\infty} nq^{2n} q^{2\ell n} \\ &= \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} nq^{2nm} \\ &= \sum_{k=1}^{\infty} \sigma_1(k) q^{2k}. \end{aligned}$$

If we recall that $F(\tau) = \pi^2/3 - 8\pi^2 \sum_{k=1}^{\infty} \sigma_1(k) q^{2k}$, we find

$$(\eta'/\eta)(\tau) = \frac{i}{4\pi} F(\tau).$$

By the chain rule, the logarithmic derivative of $\eta(-1/\tau)$ is $\tau^{-2}(\eta'/\eta)(-1/\tau)$, and using property (a), we see that the logarithmic derivative of $\eta(-1/\tau)$

equals $(i/4\pi)\tilde{F}(\tau)$. Therefore, taking the logarithmic derivative of the functional equation for η we find

$$\frac{i}{4\pi}\tilde{F}(\tau) = \frac{1}{2\tau} + \frac{i}{4\pi}F(\tau),$$

and this gives $\tilde{F}(\tau) = -2\pi i/\tau + F(\tau)$, as desired.

Finally, (c) is a consequence of (a) and (b).

To prove the transformation formula (ii) for E_2^* under $\tau \mapsto -1/\tau$, we begin with

$$E_2^*(\tau) = F(\tau/2) - 4F(2\tau).$$

Then

$$\begin{aligned} E_2^*(-1/\tau) &= F(-1/(2\tau)) - 4F(-2/\tau) \\ &= [4\tau^2 F(2\tau) - 4\pi i\tau] - 4[(\tau/2)^2 F(\tau/2) - \pi i\tau] \\ &= 4\tau^2 F(2\tau) - 4(\tau^2/4)F(\tau/2) \\ &= -\tau^2(F(\tau/2) - 4F(2\tau)) \\ &= -\tau^2 E_2^*(\tau), \end{aligned}$$

as desired. To prove the third property recall that

$$F(\tau) = \frac{\pi^2}{3} - 8\pi^2 \sum_{k=1}^{\infty} \sigma_1(k) e^{2\pi i k \tau},$$

where the sum goes to 0 as $\text{Im}(\tau) \rightarrow \infty$. Then, if we use the fact that

$$E_2^*(\tau) = F(\tau/2) - 4F(2\tau),$$

we conclude that $E_2^*(\tau) \rightarrow -\pi^2$ as $\text{Im}(\tau) \rightarrow \infty$.

To prove the final property, we begin by showing that

$$(11) \quad E_2^*(1 - 1/\tau) = \tau^2 \left[F\left(\frac{\tau-1}{2}\right) - F(\tau/2) \right].$$

From the transformation formulas for F we have

$$\begin{aligned} F(1/2 - 1/2\tau) &= F\left(\frac{\tau-1}{2\tau}\right) \\ &= \left(\frac{2\tau}{\tau-1}\right)^2 F\left(\frac{2\tau}{1-\tau}\right) - 2\pi i \frac{2\tau}{1-\tau}, \end{aligned}$$

and

$$\begin{aligned} F\left(\frac{2\tau}{1-\tau}\right) &= F(-2 + 2/(1-\tau)) \\ &= F(2/(1-\tau)) \\ &= \left(\frac{1-\tau}{2}\right)^2 F\left(\frac{\tau-1}{2}\right) - 2\pi i \left(\frac{\tau-1}{2}\right). \end{aligned}$$

Hence,

$$F(1/2 - 1/2\tau) = \tau^2 F\left(\frac{\tau-1}{2}\right) - \frac{2\pi i 2\tau}{1-\tau} - 2\pi i \frac{(2\tau)^2}{(\tau-1)^2} \left(\frac{\tau-1}{2}\right).$$

But $F(2 - 2/\tau) = F(-2/\tau) = (\tau^2/4)F(\tau/2) - 2\pi i \tau/2$, and hence

$$\begin{aligned} E_2^*(1 - 1/\tau) &= F(1/2 - 1/2\tau) - 4F(2 - 2/\tau) \\ &= \tau^2 \left[F\left(\frac{\tau-1}{2}\right) - F(\tau/2) \right] - 2\pi i \left(\frac{2\tau}{1-\tau} + \frac{2\tau^2}{\tau-1} \right) + 4\pi i \tau \\ &= \tau^2 \left[F\left(\frac{\tau-1}{2}\right) - F(\tau/2) \right]. \end{aligned}$$

This proves (11). Then, the last property follows from it and the fact that

$$F(\tau) = \frac{\pi^2}{3} - 8\pi^2 \sum_{k=1}^{\infty} \sigma_1(k) e^{2\pi i k \tau}.$$

Thus Proposition 3.8 is proved.

We can now conclude the proof of the four-squares theorem by considering the quotient $f(\tau) = E_2^*(\tau)/\theta(\tau)^4$, and applying Theorem 3.4, as in the two-squares theorem. Recall $\theta(\tau)^4 \rightarrow 1$ and $\theta(1 - 1/\tau)^4 \sim 16\tau^2 e^{\pi i \tau}$, as $\text{Im}(\tau) \rightarrow \infty$. The result is that $f(\tau)$ is a constant, which equals $-\pi^2$ by Proposition 3.8. This completes the proof of the four-squares theorem.

4 Exercises

1. Prove that

$$\frac{(\Theta'(z|\tau))^2 - \Theta(z|\tau)\Theta''(z|\tau)}{\Theta(z|\tau)^2} = \wp_\tau(z - 1/2 - \tau/2) + c_\tau,$$

where c_τ can be expressed in terms of the first two derivatives of $\Theta(z|\tau)$, with respect to z , at $z = 1/2 + \tau/2$. Compare this formula with the result in Exercise 5 in the previous chapter.